Green's theorem

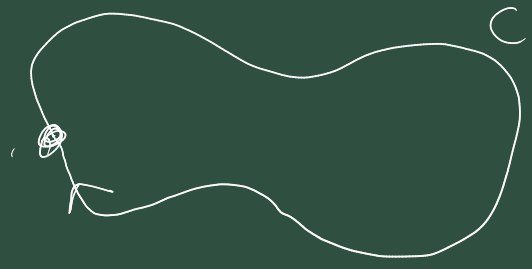
This links together the two types of integral we've seen, path integrals of lin. diff. forms, and double integrals over regions of the plane.

Closed path integrals



$$\int_S L$$

a closed curve is one whose start and end points are the same.



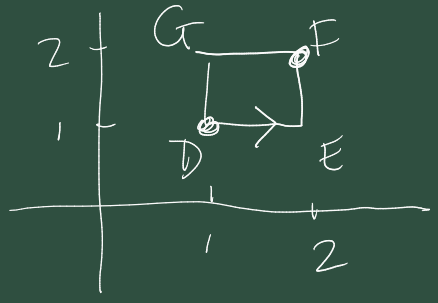
$$\oint_C L$$

a decoration on $\int$ to indicate that the path/curve C is "closed".

Example 3.8.3

Evaluate $$I = \oint_C 3x^2y^2\,dx + 2xy\,dy$$

where C is the square DEFG.



Use the decomposition.

$C = DE + EF + FG + GD$

$DE: y=1, dy=0, x: 1 \to 2.$

$EF: x=2, dx=0, y: 1 \to 2.$

$FG: y=2, dy=0 \quad x: 2 \to 1$

$GD: x=1, dx=0, y: 2 \to 1.$

$$I = \oint_C L = \int_{DE} L + \int_{EF} L + \int_{FG} L + \int_{GD} L$$

$$= \int_1^2 3x^2\,dx + \int_1^2 4y\,dy + \int_2^1 12x^2\,dx + \int_2^1 2y\,dy$$

$$= x^3\Big]_1^2 + 2y^2\Big|_1^2 + 4x^3\Big|_2^1 + y^2\Big|_2^1$$

$$= 8-1 + 8-2 + 4-32 + 1-4$$

$$= 7 + 6 - 28 - 3$$

$$= -18$$

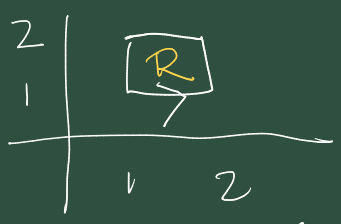
Example 3.8.10

So for $P(x,y) = 3x^2y^2$, $Q(x,y) = 2xy$

Green's Theorem claims that

$$-18 = \oint_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

where R is the interior of the square.



(where C is traversed counter-clockwise).

The double integral on the RHS of Green's theorem is

$$\iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

$$= \int_1^2 \left(\int_1^2 \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \right) dy$$

$$= \int_1^2 \left(\int_1^2 (2y - 6x^2 y)\, dx \right) dy$$

$$= \int_1^2 \left[2xy - 2x^3 y \right]_{x=1}^{x=2} dy$$

$$= \int_1^2 \Big((4y - 16y) - (2y - 2y) \Big) dy$$

$$= \int_1^2 (-12y)\, dy$$

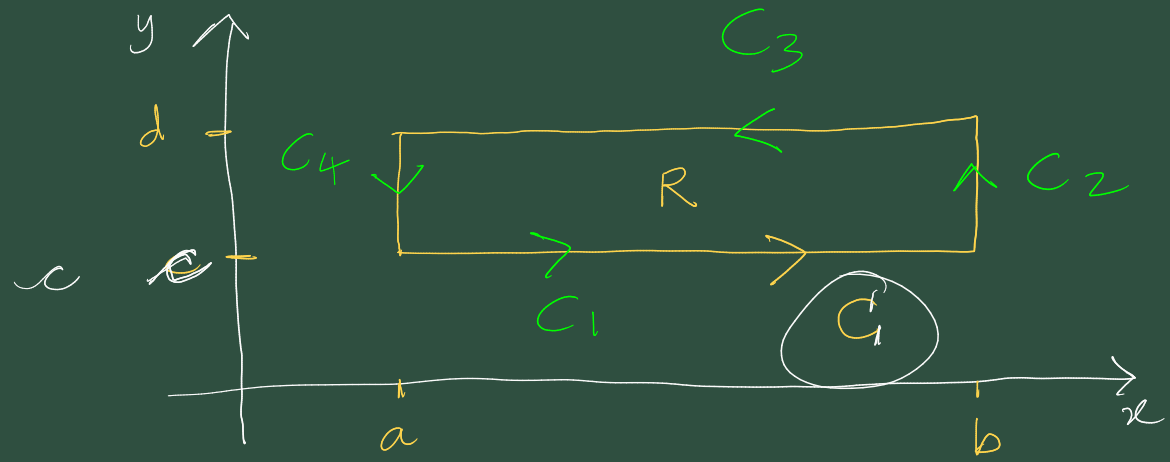
$$= \left[-6y^2 \right]_1^2$$

$$= -24 + 6 = \underline{\underline{-18}},$$ as expected from Green's theorem.

Proof of Green's theorem

Why is it true. We can explicitly/directly prove it in the simple case of rectangular regions aligned with x, y axes, and then use a nice limiting arguments for the general case.

$$C = C_1 + C_2 + C_3 + C_4$$



Consider the integral

$$\oint_C P(x,y)\,dx + Q(x,y)\,dy$$

$$= \int_{C_1} \text{—} + \int_{C_2} \text{—} + \int_{C_3} \text{—} + \int_{C_4} \text{—}$$

C_1: $y = c$, $dy = 0$, $x: a \longmapsto b$

C_2: $x = b$, $dx = 0$, $y: c \longmapsto d$

C_3: $y = d$, $dy = 0$, $x: b \longmapsto a$

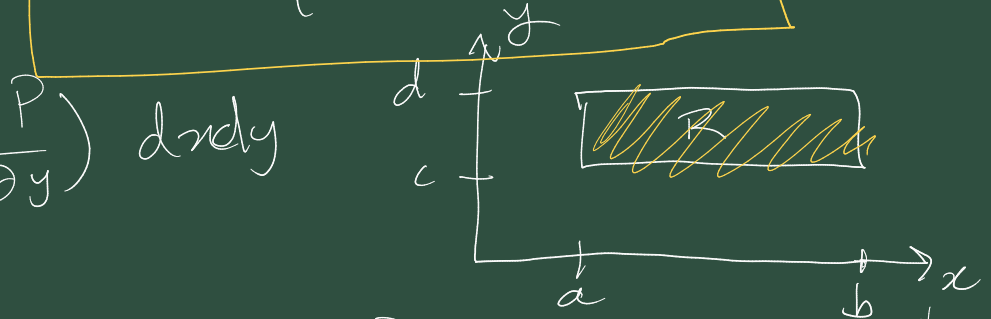
C_4: $x = a$, $dx = 0$, $y: d \longmapsto c$.

LHS of Green's theorem

$$= \int_a^b P(x,c)\,dx + \int_c^d Q(b,y)\,dy + \int_b^a P(x,d)\,dx$$

$+\int_d^c Q(a,y)\,dy$, can't develop this further as P, Q are unknown.

So consider the RHS of Green's Th.



$$= \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

$$= \iint_R \frac{\partial Q}{\partial x}\,dx\,dy - \iint_R \frac{\partial P}{\partial y}\,dx\,dy \quad \text{by linearity.}$$

$dx\,dy \rightarrow dy\,dx$

$$= \int_c^d \left(\int_a^b \frac{\partial Q}{\partial x}\,dx\right) dy - \int_a^b \left(\int_c^d \frac{\partial P}{\partial y}\,dy\right) dx$$

$$= \int_c^d \Big[Q(x,y)\Big]_{x=a}^{x=b}\,dy$$

$$- \int_a^b \Big[P(x,y)\Big]_{y=c}^{y=d}\,dx$$

$$= \int_c^d \Big(Q(b,y) - Q(a,y)\Big)\,dy$$

$$- \int_a^b \Big(P(x,d) - P(x,c)\Big)\,dx$$

$$= \int_c^d Q(b,y)\,dy + \int_d^c Q(a,y)\,dy + \int_b^a P(x,d)\,dx$$

$$+\int_a^b P(x,c)\,dx$$

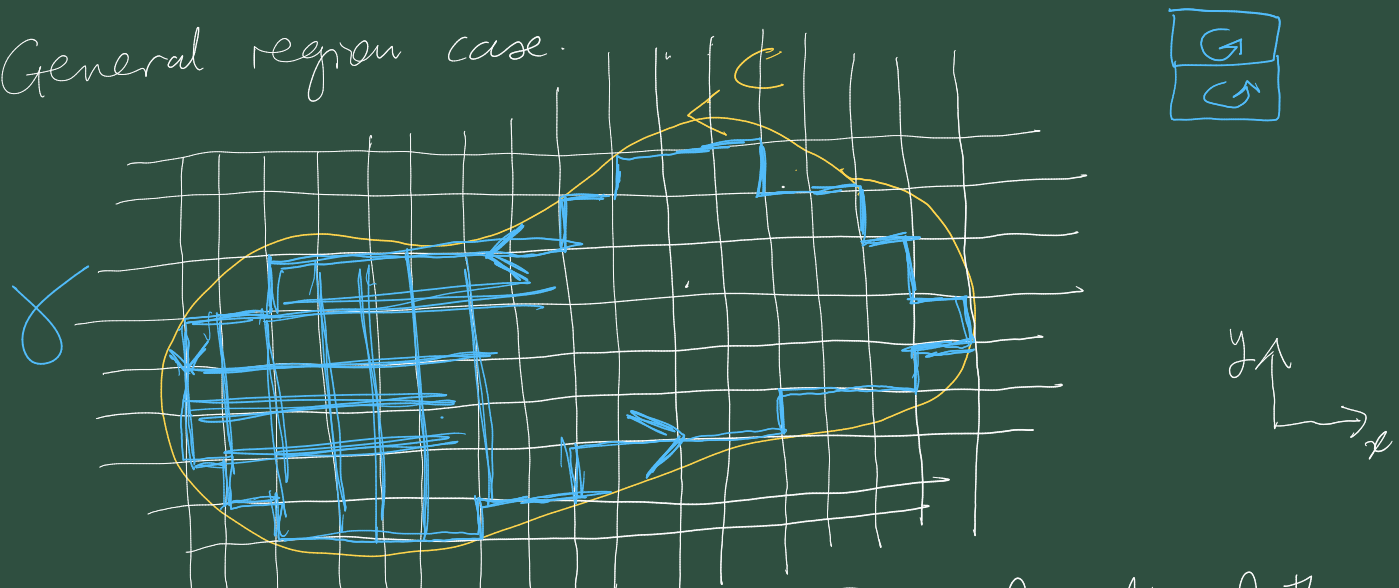
= LHS of Green's theorem we found above.

So this proves Green's theorem for these aligned rectangles.

(continue at 12:08)



General region case.



Consider an arbitrary region R, and ruling of the plane provided by partitions of the x, y axes.

Consider the union of rectangles lying wholly within R,

$$R \approx \bigcup \square$$

moreover as the sizes of rectangles $\to 0$ in the limit this approximation will

become exact

$$R = \lim_{\substack{\text{size } \square \\ \square \to 0}} \bigcup_{\square} \square$$

RHS of GT.

$$= \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy$$

$$\approx \iint_{\bigcup \square} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy$$

$$= \sum_{\square} \iint_{\square} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy$$

$$= \sum_{\square} \oint_{\partial \square} P\,dx + Q\,dy$$

$$= \oint_{\gamma} P\,dx + Q\,dy$$

$$\approx \oint_{C} P\,dx + Q\,dy$$

$$= \text{RHS of GT.}$$

as γ approximates C:

Notice that we integrate across each shared edge twice, once in one direction, one in the opposite direction. So all sub-integrals cancel out except for those edges that are not shared edges. These make up the boundary of the union of rectangles within R, call this γ.

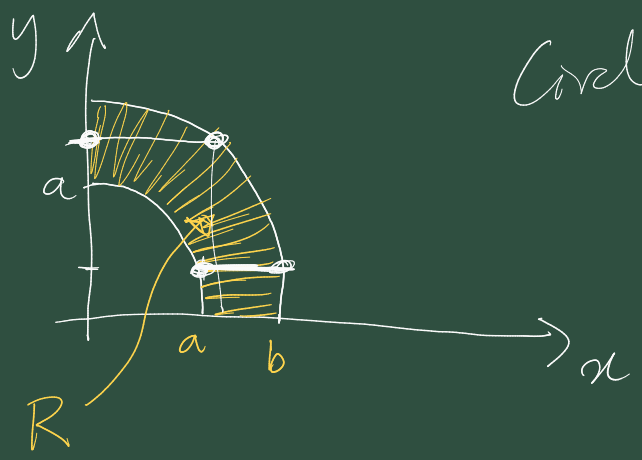
But in the limit $\gamma \to C$. Taking the limit of these approximations will prove Green's Theorem

for C and R.

Changing coordinate systems in multi-integrals.

Example Consider the integral

$$I = \iint_R \frac{1}{x^2+y^2}\,dx\,dy$$ where R is the annular region in the upper-right quadrant between circles of radius a and b.



Circles. $x^2+y^2=a^2$

$x^2+y^2=b^2$

$$I = \iint_R \frac{1}{x^2+y^2}\,dx\,dy$$

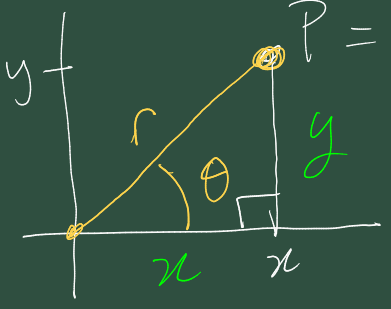
$$= \int_0^b \left(\int_{?}^{\sqrt{b^2-y^2}} \frac{1}{x^2+y^2}\,dx \right) dy$$

$$= \int_0^a \left(\int_{\sqrt{a^2-y^2}}^{\sqrt{b^2-y^2}} \frac{1}{x^2+y^2}\,dx \right) dy$$

$$+ \int_a^b \int_0^{\sqrt{b^2-y^2}} \frac{1}{x^2+y^2}\,dx\,dy$$

= YUKES; Problem: Cartesian coordinates are not suited to this task.

Aspects of this integral are very circular. Which suggests using Polar coordinates (r, θ), which are well suited to describing circular things.



$$r^2 = x^2 + y^2$$

$$x = r\cos\theta$$

$$y = r\sin\theta$$

To transform double integrals wrt Cartesian coordinates to integrals in polar coordinates.

equivalent

$$\iint_R f \, dx\,dy = \int\int f \left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| dr\,d\theta$$

where $\left|\frac{\partial(x,y)}{\partial(r,\theta)}\right|$ is the Jacobian determinant of the transformation.

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= \cos\theta\, r\cos\theta + \sin\theta\, r\sin\theta$$

$$= r(\cos^2\theta + \sin^2\theta)$$

$$= r$$

So $dx\,dy = r\,dr\,d\theta$. Returning to I.

$$I = \iint_R \frac{1}{x^2+y^2}\,dx\,dy$$

$$= \iint_R \frac{1}{r^2}\, r\,dr\,d\theta$$



$$= \iint_R \frac{1}{r}\,dr\,d\theta$$

$$= \int_0^{\pi/2} \left(\int_a^b \frac{1}{r}\,dr \right) d\theta$$

$$= \int_0^{\pi/2} \left[\ln(r) \right]_a^b d\theta$$

$$= \int_0^{\pi/2} \left(\ln(b) - \ln(a) \right) d\theta$$

$$= \int_0^{\pi/2} \ln\left(\frac{b}{a}\right) d\theta$$

$$= \ln\left(\frac{b}{a}\right) \int_0^{\pi/2} d\theta$$

$$= \frac{\pi}{2} \ln\left(\frac{b}{a}\right)$$